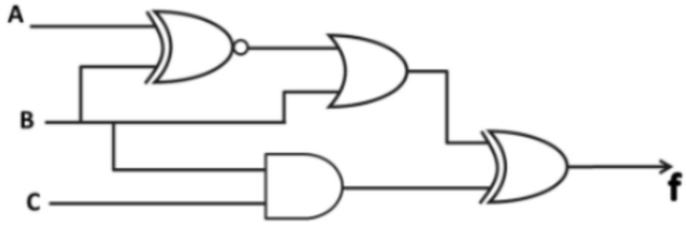
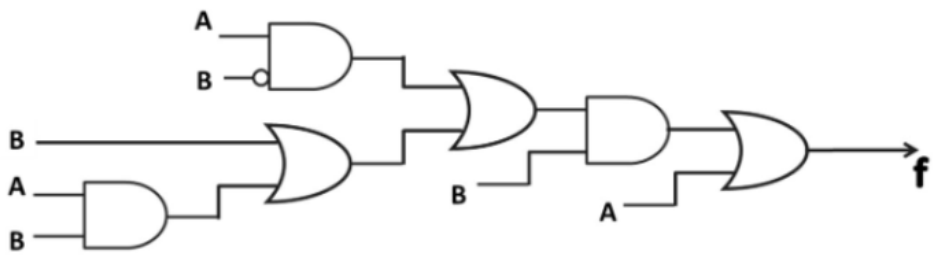
Boolean Algebra- Assignment

1. Derive the output expression ‘f’ for the below circuits and draw the simplified circuits.

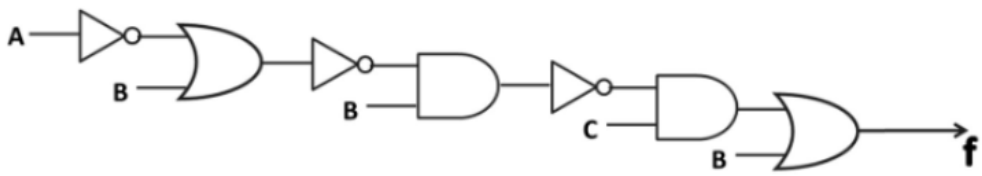
a)



b)



c)



2.Minimize the following Boolean Expressions using Boolean postulates.

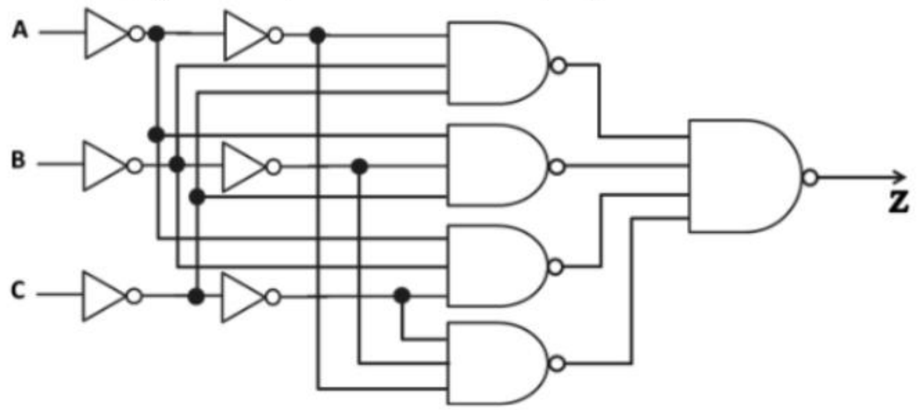
a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

d) $\overline{x_1}\,\overline{x_3}\,\overline{x_4} + x_3x_4 + \overline{x_1}\,\overline{x_2}x_4 + x_1x_2\overline{x_3}x_4 + x_2$

3.Which single gate can represent the functionality of given circuit?



4. Minimize the following Boolean Expressions using K - map

a) $f(a,b,c,d) = \sum_m(4,6,8,10,11,12,15) + \sum_d(3,5,7,9)$

b) $f(a,b,c,d) = \prod_M(0,2,4,6,7,9).\prod_D(10,11)$